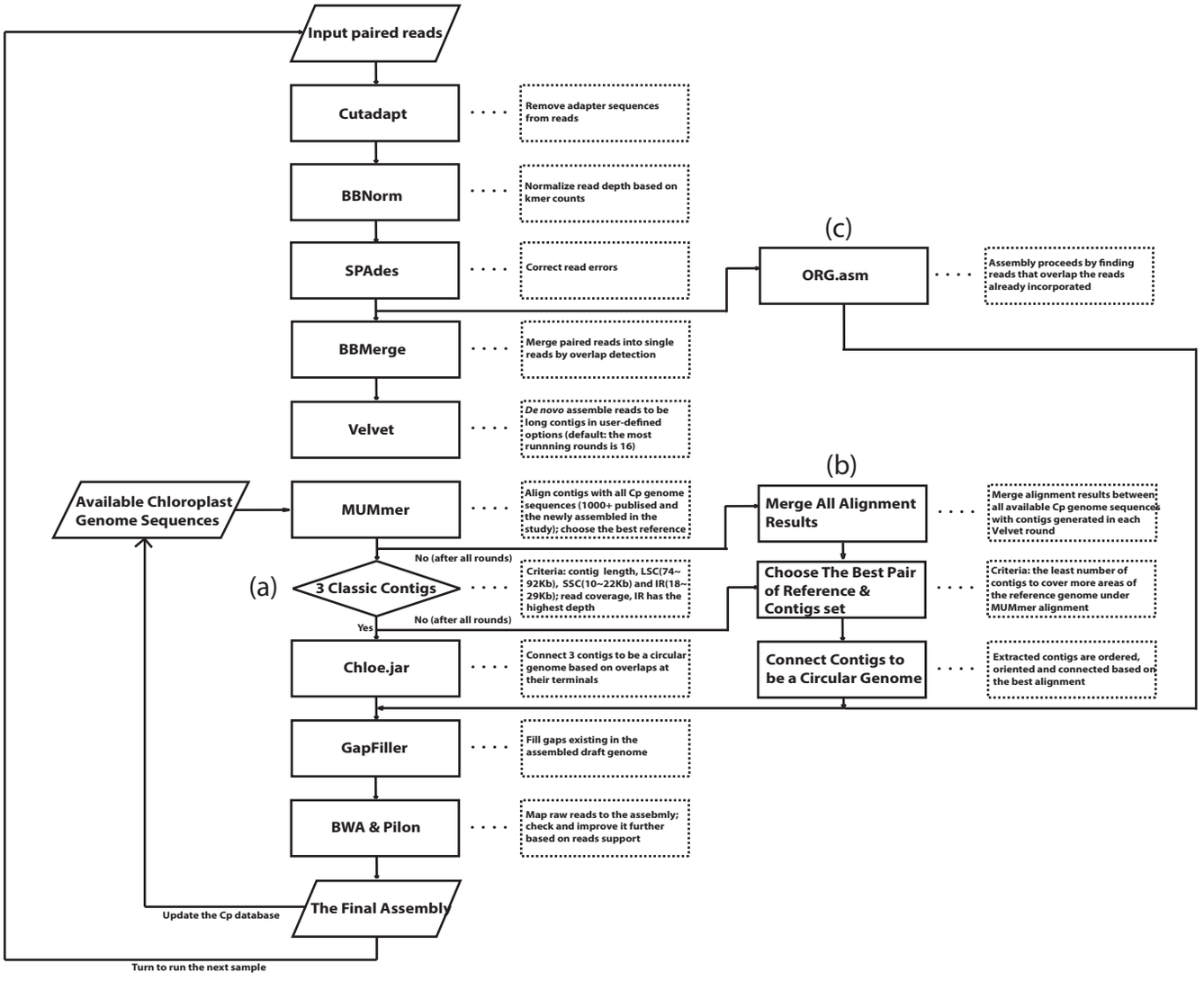
Input paired reads
Cutadapt
Remove adapter sequences from reads
BBNorm
Normalize read depth based on kmer counts
SPAdes
Correct read errors
BBMerge
Merge paired reads into single reads by overlap detection
Velvet
De novo assemble reads to be long contigs in user-defined options (default: the most runnning rounds is 16)
Available Chloroplast Genome Sequences
MUMmer
Align contigs with all Cp genome sequences (1000+ publised and the newly assembled in the study); choose the best reference
No (after all rounds)
(a)
3 Classic Contigs
Criteria: contig length, LSC(74~92Kb), SSC(10~22Kb) and IR(18~29Kb); read coverage, IR has the highest depth
Yes
No (after all rounds)
Chloe.jar
Connect 3 contigs to be a circular genome based on overlaps at their terminals
GapFiller
Fill gaps existing in the assembled draft genome
BWA & Pilon
Map raw reads to the assebmly; check and improve it further based on reads support
The Final Assembly
Update the Cp database
Turn to run the next sample
(c)
ORG.asm
Assembly proceeds by finding reads that overlap the reads already incorporated
(b)
Merge All Alignment Results
Merge alignment results between all available Cp genome sequences with contigs generated in each Velvet round
Choose The Best Pair of Reference & Contigs set
Criteria: the least number of contigs to cover more areas of the reference genome under MUMmer alignment
Connect Contigs to be a Circular Genome
Extracted contigs are ordered, oriented and connected based on the best alignment